

H41 Variance-Ratio Trending Momentum

Intraday momentum on NQ futures, taken only when the market is in a trending (high variance-ratio) regime: research report (cohort member #19, C91756637ca7282e7)

Prepared for the investment committee · 23 September 2026 · data through 10 August 2026 · all figures per 1 NQ contract (\$20/point) unless stated

Recommendation: FIRST CANDIDATE FOR FORWARD TESTING. Run a small, fixed-risk pilot next to the true-forward paper record. Do not trade full size yet. H41 is the only one of the 25 frozen strategies with clear profits in every test after its selection period, on a large sample. Untouched holdout 2023-25: +\$143.4k, PF 1.31, t 2.43, null p 0.010. 2026 to date: +\$56.7k, PF 1.45. The edge is thin, though: PF 1.2-1.4, a 30% win rate, and most of the profit comes from a few big trend days. It also barely made money from 2010 to 2016. Plan for roughly half the recent edge.

Executive summary

- **What it is.** It buys (or sells short) NQ when the last 15 minutes moved at least 1.5 standard deviations up (or down). It only does this when the previous two hours were *trending*: the variance ratio VR(5) is at least 1.2, meaning moves have been building on each other rather than reversing. Stop is 1× ATR. There is no target, no trailing stop and no break-even, and it is always flat by 16:00 New York. **About 5 trades a week** (about 290 a year), split evenly between long and short.
- **Evidence ladder.** Discovery 2010-18: +\$31.5k, 1,852 trades, PF 1.14, t 1.66. Confirmation 2019-22: +\$99.8k, 1,207 trades, PF 1.18, t 1.92, null p 0.039, tier PROMISING. **Untouched holdout 2023-25: +\$143.4k, 843 trades, PF 1.31, t 2.43, null p 0.010 (the best in the cohort), SUPPORTED.** 2026 audit (Jan-Aug): +\$56.7k, 158 trades, PF 1.45, SUPPORTED.
- **After the freeze** (2023-Aug 2026, data never used to pick it): 1,001 trades, **PF 1.34**, +\$200.1k, +281R, +0.28R per trade, t 3.2 in R. The stress-cost version is still +\$188.1k. Every year was positive. 2023 was the weakest year: +\$4.8k, +24R.
- **How it makes money.** About two-thirds of trades are stopped out near -1R, typically within about 25 minutes. The remaining third run to the close and average +2.7R. Average winner ≈ 3× average loser, win rate 29-35%.
- **Main weaknesses.** (1) The edge did not exist in 2010-2016: -79R over seven years and a 7½-year drawdown. It appeared from 2017. (2) It is thin and depends on its tail: the 10 best trades made 80% of holdout net profit. (3) Long losing runs: 40 losses in a row in 2019-22, and a \$48.4k drawdown on one contract. (4) After correcting for the whole search it is not significant (family-wise p 0.35), so the clean evidence is the holdout alone. (5) Daily correlation is 0.64 with H20 and 0.65 with the rest of the cohort, so it is part of one shared momentum bet.
- **Forward view.** We expect it to stay profitable but weaker than 2023-26. Our planning case is +0.12R per trade (about 40% of the post-freeze average), which gives a median of about +33R a year and roughly a 1-in-5 chance of a losing year. A return to a 2010-16-style regime, where the edge disappears, is the main risk. Pre-agreed kill rules are on the last pages.

Headline results by research stage

Period	Trades	Per week	Net P&L	PF	Win %	Net R	Avg R	Max DD	Verdict
Discovery 2010-18	1,852	4.1	\$31,467	1.14	27.9%	+127.3	+0.069	\$15,221	PRIMARY
Confirmation 2019-22	1,207	5.8	\$99,842	1.18	29.3%	+214.3	+0.178	\$48,427	PROMISING
Holdout 2023-25	843	5.4	\$143,363	1.31	30.5%	+218.5	+0.259	\$17,936	SUPPORTED
2026 audit (Jan-Aug)	158	4.9	\$56,733	1.45	34.8%	+62.9	+0.398	\$17,573	SUPPORTED
2024-2025 only	564	5.4	\$138,559	1.41	31.9%	+194.2	+0.344	\$17,936	-
All post-freeze 2023-26	1,001	5.3	\$200,096	1.34	31.2%	+281.4	+0.281	\$21,361	-

Stages before the freeze (discovery, confirmation) were used to select the strategy; 2023 onward was never seen before the cohort was frozen on 22 Sep 2026 23:21 UTC. Verdicts are the campaign's preregistered classifications. $R = \text{P\&L}$ divided by the trade's own initial risk.

1. How this strategy was found and tested

The strategy came out of the QuantLab V3 research campaign: 53 preregistered hypothesis families (33,077 specifications) were searched on 2010–2018 NQ/ES 1-minute data. Survivors were frozen and re-tested unchanged on 2019–2022 against 100 synthetic ‘no-edge’ market worlds, clustered into distinct behaviours, and 25 were selected into a frozen cohort. The cohort was then run once on the untouched 2023–2025 holdout, and once on 2026 data to 10 Aug 2026. Nothing was changed after any later stage was seen; that is what makes the post-2022 numbers honest out-of-sample evidence.

Stage	Dates	Role	What could change
Discovery	2010-06-07 → 2018-12-31	search (in-sample)	everything: rules, grid, management
Confirmation	2019 → 2022	out-of-sample test vs null worlds	only which frozen candidates are kept
Final cohort freeze	22 Sep 2026 23:21 UTC	spec hash f5666dac...	nothing after this point
Holdout	2023 → 2025	untouched out-of-sample	nothing (report only)
2026 audit	2026-01-01 → 2026-08-10	secondary, low power	nothing (report only)
True forward	from 2026-09-24	live-market paper record	nothing (forward/run_forward.py)

Reproducibility. For this report every trade from 2010 to 2026 was regenerated with the frozen specification, the stage-gated data loader and the campaign engine. Trade counts and net P&L match the sealed campaign records exactly in all four periods for every strategy covered here.

Why R-multiples matter here. NQ traded near 2,000 in 2010 and near 27,000 in 2026, and the stop is volatility-based, so one trade's dollar risk ranged from roughly \$50 to over \$2,000 on one contract. Dollar totals are therefore dominated by recent, high-volatility years, and a single contract is a very different bet in 2012 than in 2026. We report both dollars (the campaign's preregistered metric) and R ($\text{P\&L} \div \text{initial risk}$), which approximates a book that risks a constant amount per trade. **The official verdicts use dollars; the R view is a descriptive, post-hoc lens and cannot overturn them.**

Hypothesis family

Preregistered hypothesis (H41). Momentum $Z(15) \geq z$, with a confluence condition: the variance ratio $VR(5)$ over the last L bars $\geq vr$ (a persistent, trending regime). Grid: $z \in \{1.0, 1.5\}$, $vr \in \{\text{none}, 1.2, 1.4\}$, $L \in \{120, 240\}$, continuation only, long / short / both. In discovery 758 specifications were tested and 119 were PRIMARY (best t 2.69). $Z(n)$ is the n -minute price change divided by that time of day's normal n -minute volatility over the previous 20 sessions, so a 1.5 means the same thing at 09:45 and at 14:00, in 2012 and in 2026.

Rationale. $VR(5)$ compares the variance of 5-minute returns with five times the variance of 1-minute returns. In a random walk it is 1.0. Above 1, minute-to-minute moves have been continuing each other: orders are being worked in one direction, and trend-following and hedging flows are building. A strong 15-minute push inside such a regime is more likely to extend into the rest of the session than the same push inside a choppy, mean-reverting market. The hypothesis says the VR filter should separate the two.

The filter does real work. In the campaign's paired ablation, adding the VR condition raised the average trade in 91% of 220 discovery pairs and in 76% of 42 confirmation pairs, while removing about 83% of trades. For this exact entry, the unfiltered version (same $z = 1.5$, no VR condition) had PF 0.99 in discovery, 1.09 in confirmation, 1.10 in the holdout and 0.93 in 2026. With the filter: 1.14 / 1.18 / 1.31 / 1.45.

Behaviour cluster. Cluster #40 (PROMISING; 13 confirmation survivors, 2 entry rules, all H41). It overlaps the campaign's big momentum cluster (#43, represented by H20): holdout daily correlation with H20 is 0.64.

2. H41 (#19, long + short): specification

Field	Frozen value
Strategy ID	C91756637ca7282e7
Entry rule ID	E73ad53ae9e456d1a
Family / version	v3_h41_vr_trending (v3.0)
Instrument	NQ (E-mini Nasdaq-100 futures, \$20 per point), front contract, 1-minute bars
Direction	both
Mode	continuation
Long trigger	the 15-minute move first exceeds +1.5 sigma units with variance ratio VR(5) over the last 120 minutes ≥ 1.2
Short trigger	the 15-minute move first falls below -1.5 sigma units with variance ratio VR(5) over the last 120 minutes ≥ 1.2
Parameters	{"L": 120, "mode": "continuation", "vr": 1.2, "z": 1.5}
Signal timing	evaluated at the close of each 1-minute bar
Entry order	market order at the open of the next 1-minute bar (same session and contract)
Entry window	09:30-15:30 New York; one position at a time; no pyramiding
Initial stop	$1.0 \times \text{ATRh}$ from entry (ATRh = ATR(14) of completed 15-minute bars); stop order, gaps through the stop fill at the bar open; rounded away from entry to the tick
Target	none (100% of the position runs)
Break-even / trailing / partials	none / none / none
Time exit	flat at the close of the last bar before 16:00 New York
Same-bar ambiguity	pessimistic (stop assumed first)
Costs (baseline)	\$14 per round trip = \$4 commission + 1 tick slippage per side; also scored at \$0 / \$9 / \$26
Cost file version	costs_v2_approved_2026-09-21#a27ed5df0c04

Statistical evidence trail

Stage	Test / statistic	Result	Verdict
Discovery 2010-18	label (trades ≥ 30 , net > 0 , $t \geq 1$)	1852 trades, PF 1.14, t 1.66	PRIMARY
Discovery	parameter robustness (share of grid neighbours positive)	1.00	
Discovery	sub-periods positive (of 3) / years positive (of 9)	1 / 4	
Confirmation 2019-22	t-stat of mean trade	1.92	
Confirmation	null p (zero-drift worlds) / (drift worlds) / max	0.039 / 0.039 / 0.039	$p \leq 0.05$
Confirmation	family-wise p (max-t in same frequency bucket)	0.347	not significant
Confirmation	parameter support / management support	1.00 / 0.38	
Confirmation	years positive (of 4); stress net	3; \$85,358	
Confirmation	evidence tier (preregistered)		PROMISING
Behaviour cluster	cluster at p 0.50 (members / entry rules)	#40 (13 / 2)	PROMISING
Holdout 2023-25	t-stat; z vs confirmation mean trade	2.43; 1.25	
Holdout	null p (100 holdout null worlds)	0.010	
Holdout	classification (preregistered)		SUPPORTED
2026 audit	t-stat; net	1.54; \$56,733	SUPPORTED

Null worlds randomly flip the sign of each minute's moves (keeping volatility, seasonality, gaps and the NQ-ES relation) so any directional edge is destroyed; p is the share of such worlds that match the real t-stat. The family-wise p corrects for having searched many candidates of similar frequency.

H41 (#19, long + short): how it trades

Signal (checked at the close of every 1-minute bar, 09:30-15:30 New York). Two conditions must hold. (1) The 15-minute move, measured in sigma units, *first* crosses +1.5 (long) or -1.5 (short). (2) VR(5) measured over the last 120 minutes is at least 1.2. When both hold, the strategy sends a market order at the open of the next minute.

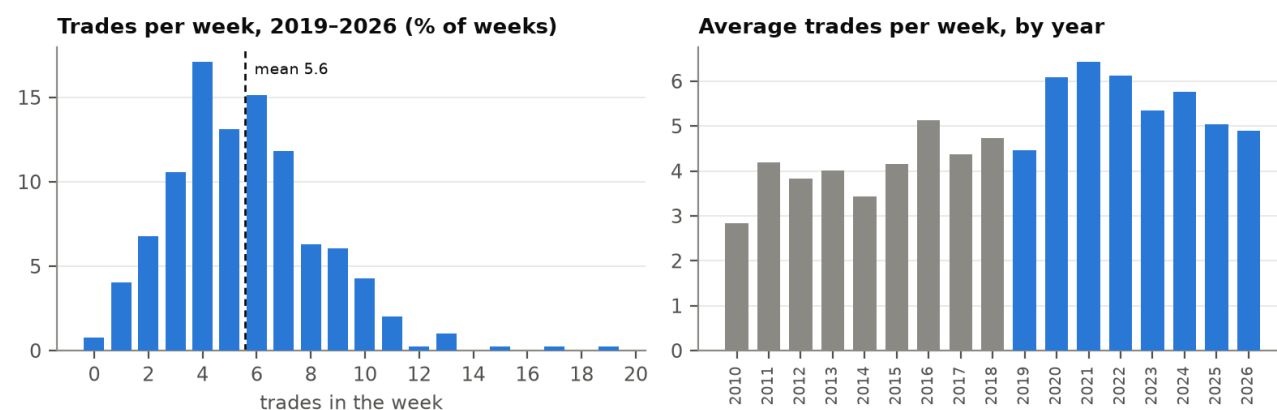
Risk and exit. The stop sits $1 \times \text{ATR}_h$ from the entry (ATR(14) of completed 15-minute bars), about 34 NQ points in 2023 and 67 in 2026, so one contract risks about \$1,350 per trade at 2026 volatility. There is no profit target, no trailing stop and no move to break-even. Anything still open is closed at the last bar before 16:00. It holds one position at a time, but can re-enter later the same day (at most 9 trades in one day historically). It never holds overnight.

What a trade looks like. Losing trades end fast: the median stopped trade lasts 24 minutes. Winning trades are held for hours: the median trade held to the close lasts 231 minutes. So the P&L profile is many small, quick losses and fewer large, slow gains. The best entries come in the first hour. Trades opened 09:30-09:59 averaged +0.43R over 2010-26 and +0.77R after 2022, while mid-morning entries were close to zero. This is descriptive only, and the rule is not changed because of it.



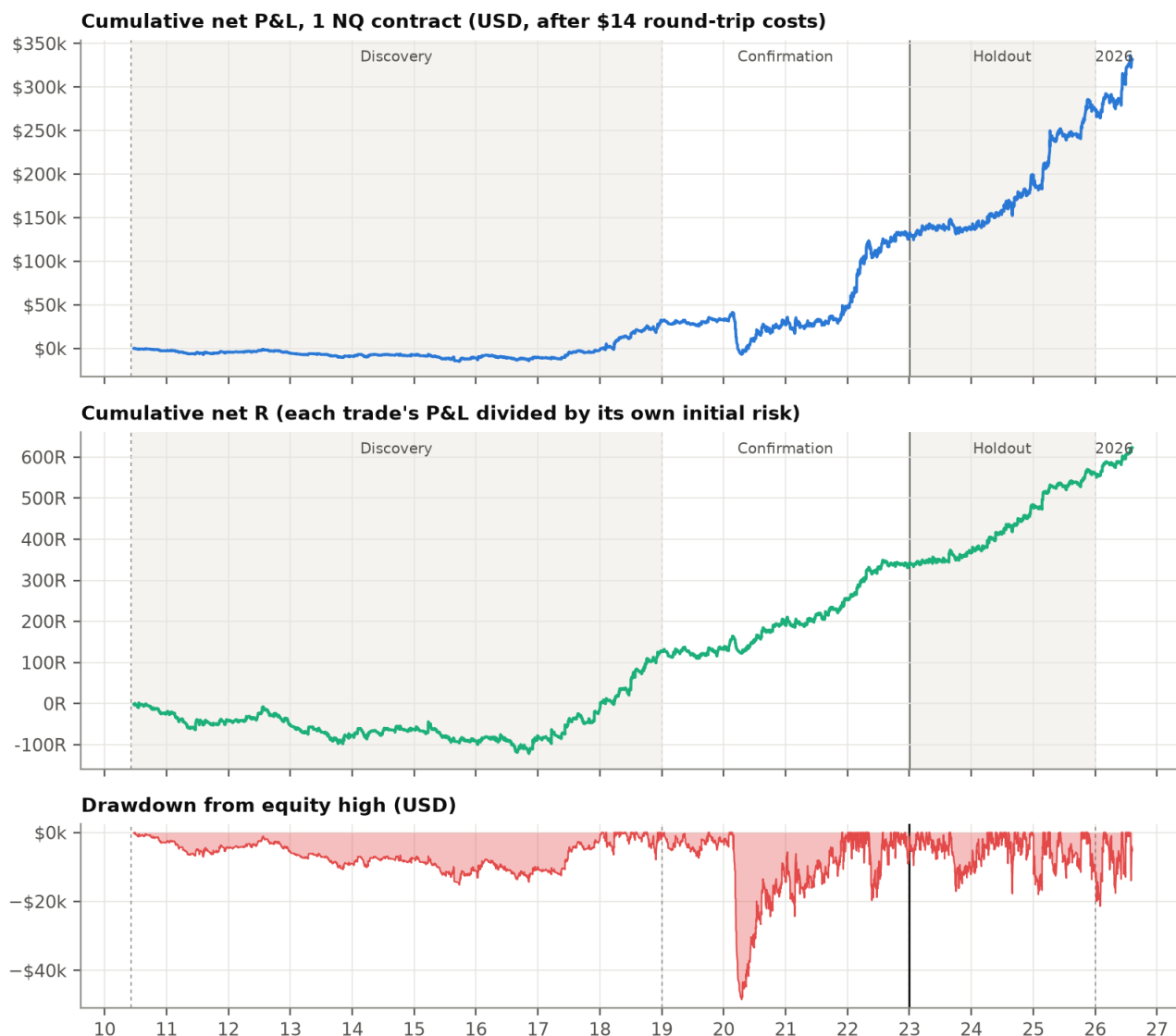
Real holdout trades, 1-minute NQ bars (grey band = bar range, line = close). Blue dotted = entry price, red dashed = the fixed stop; nothing else is managed.

How often it trades



Frequency (2019-Aug 2026)	Value
Average trades per week	5.6
Median trades per week	5
Weeks with no trade	0.8%
Weeks with 10 or more trades	8.3%
Busiest week	19 trades
Average trades per year	290

H41 (#19, long + short): performance



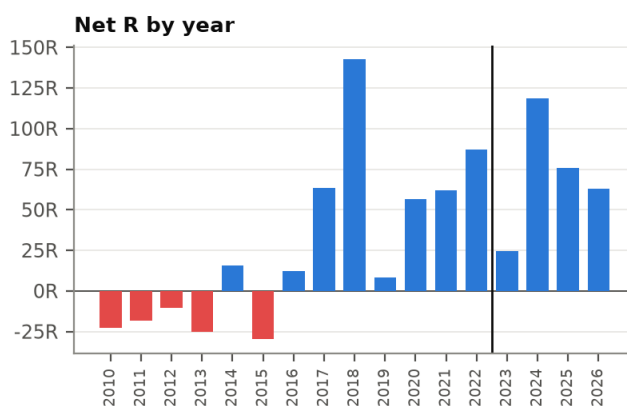
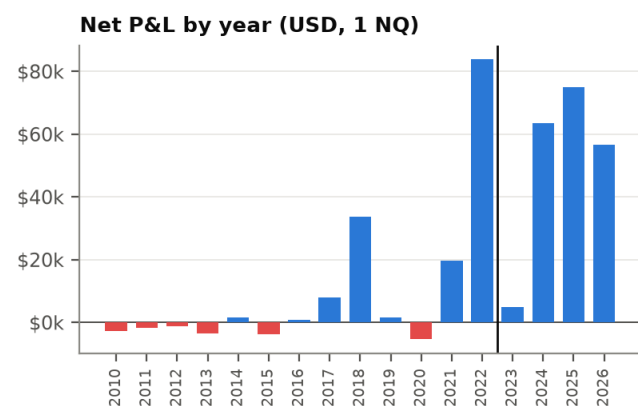
Shaded bands mark the research stages; the solid black line is the start of data never seen before the freeze (1 Jan 2023).

H41 (#19, long + short): full statistics by period

Metric	Discovery 2010-18	Confirm. 2019-22	Holdout 2023-25	2026 Jan-Aug	2024-25	Post-freeze 2023-26	Full 2010-26
Trades	1,852	1,207	843	158	564	1,001	4,060
Trades / year	216	302	281	256	282	277	251
Trades / week (all weeks)	4.1	5.8	5.4	4.9	5.4	5.3	4.8
Weeks with at least 1 trade	94%	100%	99%	100%	99%	99%	97%
Trades / week (active weeks)	4.4	5.8	5.4	4.9	5.5	5.4	5.0
Days with a trade (of sessions)	51%	68%	63%	62%	62%	63%	58%
Max trades in one day	6	9	7	6	7	7	9
Long share	48%	48%	49%	47%	48%	48%	48%
Net P&L (baseline \$14)	\$31,467	\$99,842	\$143,363	\$56,733	\$138,559	\$200,096	\$331,405
Gross P&L (no costs)	\$57,395	\$116,740	\$155,165	\$58,945	\$146,455	\$214,110	\$388,245
Net P&L, moderate \$9	\$40,727	\$105,877	\$147,578	\$57,523	\$141,379	\$205,101	\$351,705
Net P&L, stress \$26	\$9,243	\$85,358	\$133,247	\$54,837	\$131,791	\$188,084	\$282,685
Profit factor (net)	1.14	1.18	1.31	1.45	1.41	1.34	1.24
Profit factor (gross)	1.28	1.22	1.34	1.47	1.44	1.37	1.29
Profit factor (stress)	1.04	1.15	1.28	1.43	1.39	1.31	1.20
Win rate	27.9%	29.3%	30.5%	34.8%	31.9%	31.2%	29.1%
Average trade	\$17	\$83	\$170	\$359	\$246	\$200	\$82
Average winner	\$492	\$1,827	\$2,379	\$3,332	\$2,642	\$2,547	\$1,434
Average loser	-\$167	-\$641	-\$799	-\$1,228	-\$878	-\$863	-\$474
Payoff ratio (avg W / avg L)	2.95	2.85	2.98	2.71	3.01	2.95	3.02
Expectancy (net R / trade)	+0.069	+0.178	+0.259	+0.398	+0.344	+0.281	+0.153
Total net R	+127.3	+214.3	+218.5	+62.9	+194.2	+281.4	+623.0
Median trade (R)	-1.08	-1.02	-1.01	-1.01	-1.01	-1.01	-1.02
t-stat of mean trade (USD)	1.66	1.92	2.43	1.54	2.55	2.88	3.73
t-stat of mean trade (R)	1.12	2.62	2.71	1.87	2.85	3.22	3.77
Sharpe, daily, annualised (USD)	0.55	0.95	1.44	1.88	1.87	1.52	0.92
Sharpe, daily, annualised (R)	0.37	1.30	1.57	2.28	2.04	1.68	0.92
Sortino (USD)	1.05	1.64	2.90	3.91	3.88	3.10	1.76
Max drawdown (USD)	\$15,221	\$48,427	\$17,936	\$17,573	\$17,936	\$21,361	\$48,427
Max drawdown (R)	123.6	42.1	25.1	14.4	18.8	25.1	123.6
Longest drawdown (calendar days)	2,752	642	217	92	119	217	2,752
Calmar (annual net / max DD)	0.24	0.52	2.67	5.24	3.87	2.59	0.42
Longest losing streak (trades)	16	40	15	10	13	15	40
Longest winning streak	4	5	3	4	3	4	5
Best trade	\$4,661	\$15,281	\$18,151	\$16,341	\$18,151	\$18,151	\$18,151
Worst trade	-\$1,309	-\$2,589	-\$4,599	-\$3,279	-\$4,599	-\$4,599	-\$4,599
Best day / worst day	\$4,661 / -\$3,512	\$15,281 / -\$8,157	\$15,801 / -\$11,088	\$16,341 / -\$6,484	\$15,801 / -\$11,088	\$16,341 / -\$11,088	\$16,341 / -\$11,088
Top-10 trades as % of net	98%	79%	80%	162%	82%	65%	41%
Positive months	47% of 103	58% of 48	64% of 36	75% of 8	62% of 24	66% of 44	54% of 195
Average hold (minutes)	105	109	112	122	115	114	108
Avg initial risk (NQ points)	8.3	33.3	42.9	67.4	47.2	46.8	25.2
Avg initial risk, 1 NQ (USD) = 1R	\$166	\$665	\$859	\$1,348	\$944	\$936	\$504
Average NQ price at entry	4,182	11,564	18,576	27,093	20,734	19,920	10,257

Sharpe and Sortino use every business day in the period (zero on no-trade days) and are scale-free. The strategy is always flat overnight.

H41 (#19, long + short): calendar



Year	Period	N	Net P&L	Gross	Stress	PF	Win %	Avg trade	Net R	Avg R	Max DD	1R (USD)
2010	Disc.	85	-\$2,675	-\$1,485	-\$3,695	0.62	28%	-\$31	-22.9	-0.269	\$3,013	\$102
2011	Disc.	219	-\$1,621	\$1,445	-\$4,249	0.92	29%	-\$7	-18.3	-0.084	\$4,695	\$130
2012	Disc.	200	-\$1,165	\$1,635	-\$3,565	0.93	28%	-\$6	-10.7	-0.053	\$4,510	\$112
2013	Disc.	209	-\$3,481	-\$555	-\$5,989	0.79	24%	-\$17	-25.3	-0.121	\$5,180	\$100
2014	Disc.	179	\$1,669	\$4,175	-\$479	1.09	29%	\$9	+15.5	+0.087	\$3,102	\$145
2015	Disc.	217	-\$3,703	-\$665	-\$6,307	0.88	27%	-\$17	-29.7	-0.137	\$8,357	\$189
2016	Disc.	268	\$898	\$4,650	-\$2,318	1.03	26%	\$3	+12.3	+0.046	\$6,575	\$167
2017	Disc.	228	\$7,853	\$11,045	\$5,117	1.35	29%	\$34	+63.6	+0.279	\$2,375	\$137
2018	Disc.	247	\$33,692	\$37,150	\$30,728	1.59	31%	\$136	+142.7	+0.578	\$7,877	\$337
2019	Conf.	233	\$1,518	\$4,780	-\$1,278	1.03	26%	\$7	+8.4	+0.036	\$7,412	\$277
2020	Conf.	318	-\$5,287	-\$835	-\$9,103	0.97	28%	-\$17	+56.8	+0.179	\$48,427	\$654
2021	Conf.	336	\$19,746	\$24,450	\$15,714	1.13	30%	\$59	+61.8	+0.184	\$18,628	\$664
2022	Conf.	320	\$83,865	\$88,345	\$80,025	1.43	33%	\$262	+87.3	+0.273	\$19,680	\$962
2023	Holdout	279	\$4,804	\$8,710	\$1,456	1.04	28%	\$17	+24.4	+0.087	\$17,055	\$686
2024	Holdout	301	\$63,531	\$67,745	\$59,919	1.41	32%	\$211	+118.6	+0.394	\$17,724	\$806
2025	Holdout	263	\$75,028	\$78,710	\$71,872	1.41	32%	\$285	+75.6	+0.287	\$17,936	\$1,102
2026	Audit	158	\$56,733	\$58,945	\$54,837	1.45	35%	\$359	+62.9	+0.398	\$17,573	\$1,348

Monthly net R, 2019-2026 (blue = gain, red = loss; blank = no data)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2019	-1.3	-12.9	+17.5	-0.9	-2.2	-8.4	-3.2	+5.2	-1.2	+9.0	-5.0	+11.7
2020	+2.8	+22.5	-33.6	-0.5	+13.0	+14.2	+6.6	+20.2	-3.7	+14.0	+1.9	-0.6
2021	-1.9	+7.4	-4.4	-0.6	+14.2	+8.8	-2.4	+5.5	+11.6	-1.8	+12.8	+12.7
2022	+11.8	+15.7	+19.3	+26.0	-8.9	+9.3	+20.2	-12.1	+4.9	+0.9	-2.6	+2.7
2023	-6.3	+7.0	+1.5	+8.6	-1.8	-1.1	+0.1	+24.3	-22.7	+7.7	+1.8	+5.4
2024	+11.9	-4.3	+8.2	+22.8	+9.6	+3.9	+15.8	-13.2	+19.9	+10.9	-0.1	+33.3
2025	-10.7	+39.4	+3.7	+9.2	+9.6	-8.2	+8.6	+1.9	-7.2	+19.3	+17.3	-7.3
2026	-5.7	+28.7	+2.6	-11.4	+12.0	+19.2	+6.4	+11.1				

H41 (#19, long + short): trade anatomy



Volatility tercile = initial stop distance as a share of price, ranked within each calendar year.

H41 (#19, long + short): breakdowns (all periods vs post-freeze 2023-26)

Direction

Direction	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Long	1957	+0.135	1.21	\$132,712	485	+0.156	1.29	\$78,460
Short	2103	+0.170	1.27	\$198,693	516	+0.399	1.38	\$121,636

Exit reason

Exit	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
16:00 flat	1332	+2.660	55.41	\$1,665,332	353	+2.670	46.05	\$777,318
Stop-loss hit	2728	-1.070	0.00	-\$1,333,927	648	-1.020	0.00	-\$577,222

Entry hour (New York)

Entry hour	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
09:00-09:59	1087	+0.434	1.48	\$148,937	253	+0.765	1.80	\$99,093
10:00-10:59	893	-0.009	1.08	\$27,308	224	+0.182	1.32	\$44,749
11:00-11:59	548	+0.015	0.96	-\$9,477	145	-0.212	0.68	-\$36,335
12:00-12:59	522	+0.162	1.31	\$59,802	130	+0.217	1.36	\$32,815
13:00-13:59	420	+0.116	1.60	\$73,415	101	+0.159	1.75	\$37,296
14:00-14:59	426	-0.032	1.12	\$16,131	109	+0.284	1.39	\$20,364
15:00-15:59	164	+0.191	1.33	\$15,289	39	+0.065	1.10	\$2,114

Weekday

Weekday	Trades (all)	Avg R (all)	PF (all)	Net (all)	Trades 23-26	Avg R 23-26	PF 23-26	Net 23-26
Mon	765	+0.309	1.38	\$86,135	181	+0.299	1.62	\$54,021
Tue	871	+0.255	1.36	\$96,191	196	+0.348	1.50	\$54,271
Wed	831	-0.001	1.03	\$8,841	194	+0.115	1.11	\$14,224
Thu	865	+0.141	1.27	\$84,660	239	+0.406	1.19	\$29,564
Fri	728	+0.059	1.21	\$55,578	191	+0.209	1.43	\$48,016

H41 (#19, long + short): robustness

Parameter neighbourhood. Every grid point of the family with the same stop and exit, both directions separately. If the frozen point is a lucky outlier, its neighbours will look much worse. Rerun post hoc for this report; purely descriptive, and it does not re-select anything.

Entry parameters	Dir.	Disc. n / PF	Conf. n / PF	Holdout n / PF	Holdout net	2026 n / PF	2026 net
z=1.0, vr=1.2, L=120	long	1510 / 0.99	996 / 1.22	729 / 1.09	\$36,524	135 / 1.75	\$83,480
z=1.0, vr=1.2, L=120	short	1478 / 1.08	1003 / 1.17	652 / 1.23	\$85,807	128 / 0.93	-\$9,277
z=1.0, vr=1.2, L=120	both	2254 / 1.12	1507 / 1.16	1015 / 1.34	\$188,170	190 / 1.63	\$89,305
z=1.5, vr=1.2, L=120	long	1137 / 1.02	742 / 1.15	556 / 1.05	\$15,511	108 / 1.48	\$46,473
z=1.5, vr=1.2, L=120	short	1126 / 1.14	745 / 1.28	500 / 1.31	\$92,705	99 / 1.07	\$7,349
z=1.5, vr=1.2, L=120 ◀ frozen	both	1852 / 1.14	1207 / 1.18	843 / 1.31	\$143,363	158 / 1.45	\$56,733
z=1.0, vr=1.4, L=120	long	507 / 0.95	388 / 1.24	285 / 1.33	\$53,570	56 / 2.05	\$47,486
z=1.0, vr=1.4, L=120	short	496 / 1.26	412 / 1.03	268 / 1.09	\$14,308	42 / 0.92	-\$3,883
z=1.0, vr=1.4, L=120	both	816 / 1.19	658 / 1.13	452 / 1.24	\$63,277	76 / 1.80	\$47,826
z=1.5, vr=1.4, L=120	long	395 / 0.95	276 / 1.22	225 / 1.36	\$47,765	44 / 2.01	\$36,654
z=1.5, vr=1.4, L=120	short	389 / 1.38	327 / 1.01	217 / 1.11	\$14,882	35 / 1.03	\$1,185
z=1.5, vr=1.4, L=120	both	674 / 1.24	522 / 1.11	385 / 1.25	\$58,505	64 / 1.74	\$38,154
z=1.0, vr=None, L=120	long	5638 / 0.94	2626 / 1.04	1975 / 1.08	\$94,195	397 / 1.03	\$12,812
z=1.0, vr=None, L=120	short	5360 / 0.94	2450 / 1.04	1896 / 0.97	-\$30,959	363 / 1.06	\$17,918
z=1.0, vr=None, L=120	both	7490 / 0.93	3407 / 1.05	2567 / 1.11	\$151,907	511 / 1.06	\$27,881
z=1.5, vr=None, L=120	long	4148 / 0.94	1974 / 1.03	1486 / 1.01	\$8,696	323 / 0.95	-\$16,422
z=1.5, vr=None, L=120	short	3840 / 1.00	1778 / 1.13	1361 / 1.05	\$42,141	285 / 0.99	-\$2,585
z=1.5, vr=None, L=120	both	5842 / 0.99	2698 / 1.09	2033 / 1.10	\$115,318	421 / 0.93	-\$27,374
z=1.0, vr=1.2, L=240	long	874 / 0.90	655 / 1.21	472 / 1.02	\$5,612	81 / 2.11	\$70,431
z=1.0, vr=1.2, L=240	short	831 / 1.20	678 / 1.04	447 / 1.14	\$34,687	77 / 0.73	-\$20,333
z=1.0, vr=1.2, L=240	both	1278 / 1.17	990 / 1.14	683 / 1.18	\$67,528	115 / 1.28	\$27,135
z=1.5, vr=1.2, L=240	long	687 / 0.89	501 / 1.20	370 / 0.92	-\$17,875	66 / 2.13	\$63,031
z=1.5, vr=1.2, L=240	short	680 / 1.18	522 / 1.08	370 / 1.14	\$31,515	67 / 0.61	-\$29,138
z=1.5, vr=1.2, L=240	both	1094 / 1.14	821 / 1.16	604 / 1.02	\$8,729	100 / 1.18	\$16,050
z=1.0, vr=1.4, L=240	long	181 / 1.03	178 / 1.03	136 / 0.64	-\$32,269	29 / 2.27	\$29,374
z=1.0, vr=1.4, L=240	short	179 / 1.40	202 / 0.98	122 / 1.37	\$26,262	28 / 1.01	\$303
z=1.0, vr=1.4, L=240	both	299 / 1.38	317 / 0.93	214 / 0.95	-\$6,401	47 / 1.79	\$30,852
z=1.5, vr=1.4, L=240	long	148 / 1.00	141 / 0.98	109 / 0.54	-\$36,291	26 / 2.34	\$29,101
z=1.5, vr=1.4, L=240	short	147 / 1.48	158 / 1.15	111 / 1.28	\$19,131	23 / 0.50	-\$12,822
z=1.5, vr=1.4, L=240	both	257 / 1.36	251 / 1.06	191 / 0.86	-\$18,594	42 / 1.36	\$13,912

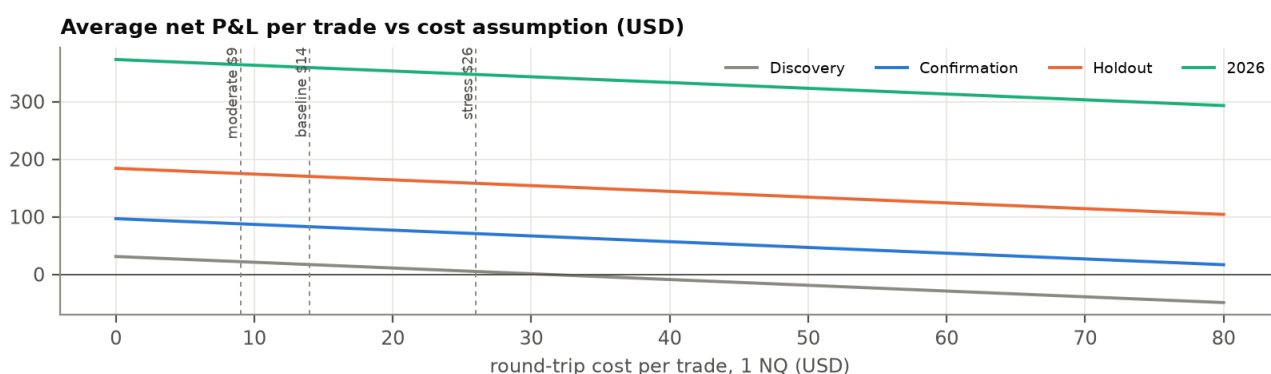
Neighbours with ≥ 20 trades and positive net: Discovery 19/30; Confirmation 27/30; Holdout 24/30; 2026 22/30

Stop / exit management. The frozen entry under every management variant from the campaign library (discovery stages A and B).

Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; single target at +1R	0.87	1.03	1386 / 1.13	\$73,796	1.08	\$14,682
1×ATR stop; single target at +1.5R	0.95	1.11	1218 / 1.05	\$31,473	1.21	\$38,520
1×ATR stop; single target at +2R	1.01	1.09	1133 / 1.11	\$62,598	1.24	\$42,569
1×ATR stop; single target at +3R	1.07	1.08	1020 / 1.09	\$52,705	1.39	\$61,890
1×ATR stop; single target at +4R	1.06	1.12	971 / 1.16	\$90,591	1.35	\$53,903
1×ATR stop; no target, flat 16:00 ◀ frozen	1.14	1.18	843 / 1.31	\$143,363	1.45	\$56,733
1×ATR stop; no target, exit after 60 min	0.96	1.16	1101 / 1.20	\$92,291	1.26	\$39,855
1.5×ATR stop; single target at +1R	0.94	1.07	1092 / 1.04	\$24,357	1.30	\$52,658
1.5×ATR stop; single target at +2R	1.03	1.06	905 / 1.03	\$15,905	1.37	\$61,400
1.5×ATR stop; no target, flat 16:00	1.12	1.13	747 / 1.11	\$60,607	1.40	\$53,410
5-min swing stop; single target at +1R	1.04	0.99	869 / 1.11	\$64,489	1.17	\$30,777
5-min swing stop; single target at +2R	1.12	1.07	733 / 1.20	\$113,298	1.51	\$79,050
5-min swing stop; no target, flat 16:00	1.18	1.15	642 / 1.24	\$120,257	1.37	\$49,713
1×ATR stop; stop to entry at +0.5R, target +2R	0.68	0.89	1360 / 0.83	-\$69,205	0.94	-\$6,575
1×ATR stop; stop to entry at +0.5R, no target	0.73	0.96	1209 / 0.89	-\$40,041	0.84	-\$14,742

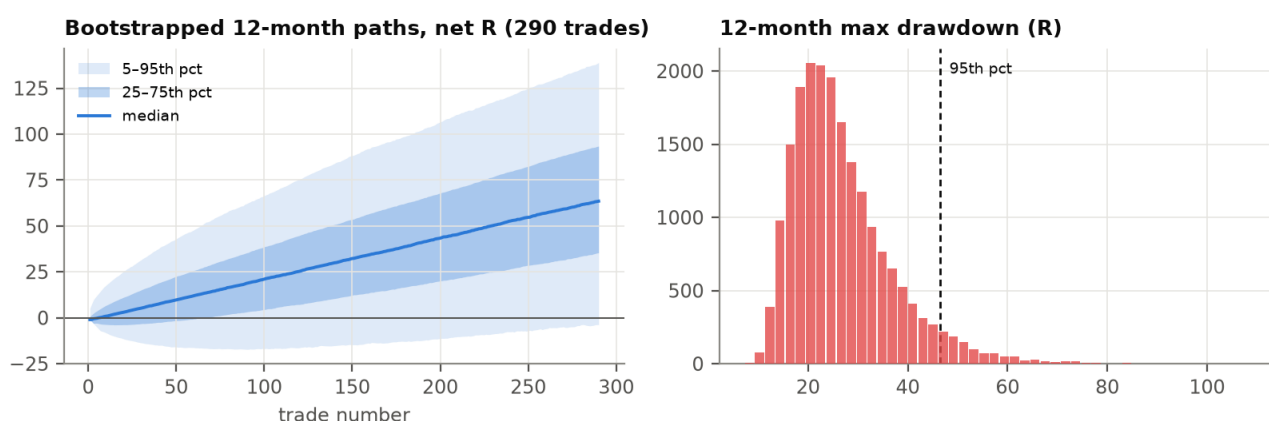
Stop / exit management	Disc. PF	Conf. PF	Holdout n / PF	Holdout net	2026 PF	2026 net
1×ATR stop; stop to entry at +1R, target +2R	0.88	1.03	1220 / 1.04	\$20,850	0.99	−\$1,712
1×ATR stop; stop to entry at +1R, no target	0.99	1.17	1004 / 1.20	\$80,714	1.07	\$8,672
1×ATR stop; stop to entry at +1.5R, target +2R	0.97	1.13	1163 / 1.07	\$41,918	1.15	\$26,250
1×ATR stop; stop to entry at +1.5R, no target	1.11	1.26	914 / 1.31	\$130,009	1.28	\$33,393
1×ATR stop; no target, 1.5×ATR trailing stop	0.98	1.15	1095 / 1.00	\$1,565	1.33	\$44,932
1×ATR stop; no target, 2×ATR trailing stop	1.07	1.19	984 / 1.10	\$48,329	1.45	\$60,467
1×ATR stop; no target, 3×ATR trailing stop	1.10	1.16	879 / 1.26	\$121,219	1.57	\$71,943
1×ATR stop; no target, trail beyond last 5-min swing	1.12	1.16	983 / 1.21	\$94,248	1.44	\$60,313
1×ATR stop; 50% at +1R, 50% at +2R	0.94	1.09	1133 / 1.16	\$70,258	1.18	\$27,544
1×ATR stop; 50% at +1R, 50% at +3R	0.98	1.06	1020 / 1.12	\$50,838	1.37	\$45,122
1×ATR stop; 50% at +1R, rest trails 2×ATR	0.97	1.15	982 / 1.14	\$56,357	1.37	\$41,704
1×ATR stop; 25% at +1R, 75% runs to 16:00	1.09	1.18	843 / 1.31	\$122,684	1.45	\$48,258
1×ATR stop; 50% at +1R then stop to entry, rest +3R	0.90	1.07	1125 / 1.09	\$40,570	1.18	\$25,473
1×ATR stop; stop to entry at +1R, 2×ATR trail	0.95	1.14	1061 / 1.05	\$19,891	1.21	\$28,190

Cost sensitivity. The line shows what the average trade would be at any round-trip cost.



Break-even round-trip cost per contract (gross average trade): Discovery \$31; Confirmation \$97; Holdout \$184; 2026 \$373. The baseline assumption is \$14.

H41 (#19, long + short): risk, sizing and forward expectations



Bootstrap of 2208 trades from 2019–Aug 2026 (all out-of-sample relative to discovery), 290 trades per simulated year, 20,000 paths, trades drawn independently. This ignores regime persistence, so real drawdowns cluster more than shown. It assumes the 2019–26 edge continues unchanged; the section 'Will it perform like this in the future?' repeats it with smaller edges.

Forward 12-month statistic (bootstrap)	In R	1 NQ at 2026 risk (1R ≈ \$1,348)	1 MNQ (1R ≈ \$135)
Median 12-month result	+63.5	\$85,551	\$8,555
5th percentile (bad year)	−3.7	−\$5,048	−\$505
95th percentile (good year)	+138.9	\$187,192	\$18,719
Probability of a losing year	6%		
Median max drawdown	24.7	\$33,270	\$3,327
95th pct max drawdown	46.4	\$62,547	\$6,255
99th pct max drawdown	59.8	\$80,617	\$8,062
Historical max drawdown 2010–26	123.6		

Suggested capital rule of thumb (not a margin requirement): hold at least the 99th-percentile 12-month drawdown plus exchange margin per unit traded. Dollar outcomes on one full NQ contract are far larger today than in most of the backtest; micro contracts (MNQ, 1/10 size) allow risk to be held near-constant per trade.

Portfolio context (25-strategy frozen cohort, 1 contract each)

Period	Corr. with rest of cohort (daily)	Median pairwise corr.	Most correlated members	Share of cohort P&L
holdout	0.65	0.14	#1 H20 both (0.64), #15 H41 short (0.53), #12 H43 both (0.40), #9 H08 short (0.36)	23%
2026	0.63	0.24	#1 H20 both (0.51), #9 H08 short (0.45), #23 H47 both (0.36), #11 H07 long (0.34)	57%

Strengths and weaknesses

Strengths

- **Positive in every test after it was selected, on a large sample.** 2,208 trades from 2019 to Aug 2026, all out-of-sample relative to discovery, each stage profitable. It improved after the freeze: holdout t 2.43 and 2026 PF 1.45, compared with t 1.92 in confirmation. Most strategies decay after selection. This one did not (so far).
- **Strongest untouched-holdout result in the cohort.** Null p 0.010 against 100 synthetic no-edge markets, the lowest of all 25 members, and SUPPORTED under the preregistered rule.
- **Not a lucky parameter.** Most VR-filtered neighbours on the grid were profitable in the holdout (counts on the robustness page). 27 of 29 alternative stop/exit schemes were profitable in the holdout and 26 of 29 in 2026. The only consistent losers are tight break-even rules, which cut the big trend days the strategy relies on.
- **The filter carries the edge.** The same momentum trigger without the VR condition is roughly break-even. That fits the economic story, and it is evidence against the profits being a quirk of the entry trigger.
- **Simple and hard to over-fit.** Two inputs, one fixed stop, time exit. No targets, trailing rules or partials to tune.
- **Cheap to execute and robust to costs.** It uses market orders in the most liquid equity-index future, during RTH only. The holdout gross average trade is far above the \$14 baseline cost. Under stress costs (2 ticks per side plus 1.5× commission) the holdout is still +\$133.2k, PF 1.28.
- **Both directions work.** Over 2010–26 longs made +264R and shorts +359R. It is not a disguised long-Nasdaq bet.
- **Frequent enough to judge quickly.** About 5 trades a week, and 99% of weeks have at least one trade. About 150 forward trades, enough for a first verdict, take roughly 7 months.

Weaknesses and risks

- **The edge was absent for seven years.** 2010–2016 was –79R, with 5 of 7 years negative, and the discovery drawdown lasted 2,752 days. Almost all discovery profit came in 2017–18 (the campaign flagged it 'regime-conditional'). Whatever drives the effect has not always been there, and it can go away again.
- **Thin edge, fat right tail.** Win rate 30%. The 10 best trades were 80% of holdout net profit. Missing a handful of trend days (a platform outage, a skipped signal) can erase a year. Execution must be fully systematic.
- **Painful losing runs.** Longest losing streak 40 trades (2019–22) and 15 in the holdout. Max drawdown on one NQ contract: \$48.4k (42R) in confirmation and \$17.9k (25R) in the holdout. The 2019–22 drawdown lasted 642 days.
- **Weak after correcting for the search.** The family-wise p is 0.35, and confirmation ranked it only PROMISING, not STRONG. Among thousands of candidates, one this good in 2019–22 could have appeared by chance. The holdout (p 0.010) is the evidence that counts, and it is only three years.
- **Selection bias in this report.** It was chosen as 'the strongest' of 25 *after* seeing the holdout and 2026. Choosing the best of 25 on those results inflates them, so forward results should be expected to come in lower.
- **Dollar risk has grown.** One full NQ contract now risks about \$1,350 per trade (2026), compared with \$665 in 2019–22 and \$166 in discovery. Dollar drawdowns in the backtest understate today's. Size in risk units (micro contracts), not contracts.
- **Correlated with the rest of the book.** Holdout daily correlation 0.64 with H20 (#1), 0.53 with the short-only H41 (#15), and 0.65 with the other 24 cohort members combined. It was 23% of cohort holdout P&L. Trading it alongside H20 is largely the same momentum bet.
- **Recent regime dependence.** 2024–25 (+\$138.6k) produced most of the holdout profit. Those were strong, trending Nasdaq years. 2023 was nearly flat (+\$4.8k, +24R).

Will it perform like this in the future?

Short answer: probably not this well. The 2023–26 numbers are the most honest evidence we have, but three things point to weaker results ahead. **Selection:** it was picked as the best of 25 after the holdout, so luck is baked in. **Mean reversion across periods:** the average trade was +0.07R in discovery, +0.18R in confirmation and +0.28R after the freeze. The long-run average (2010–26) is +0.15R. **Regime:** the edge was missing from 2010 to 2016. The table resamples its own trades under five assumptions about the true average trade.

Scenario for the next 12 months	Avg R / trade	PF (R)	Median 12-mo	Bad year (5th pct)	Good year (95th)	P(losing year)	95th pct max DD
As in the post-freeze record (2023–26)	+0.281	1.42	+80R	+12R	+155R	2%	40R
As in confirmation (2019–22)	+0.178	1.23	+50R	–18R	+125R	11%	53R
Planning case: edge roughly halved	+0.120	1.15	+33R	–34R	+108R	21%	63R
Weak edge	+0.050	1.05	+13R	–55R	+88R	39%	77R
Edge gone (like 2010–16)	+0.000	0.99	–2R	–69R	+73R	52%	88R

Each row resamples the 2019–2026 trades (290 trades per simulated year) with the average trade shifted to the scenario value. In dollars, multiply R by about \$1,348 for 1 NQ or \$135 for 1 MNQ at 2026 volatility. The highlighted row is the planning case this report recommends.

Our planning case is the highlighted row: about +0.12R per trade. That is below confirmation and about 40% of the post-freeze average. In it, the strategy still makes money in most years, but roughly one year in five is a loss, and a 12-month drawdown of 40–60R should be expected at some point. If the true edge is what 2023–26 showed, the pilot will look excellent. The plan does not depend on that.

What would make us more confident: 150+ forward trades at $\geq +0.08R$, a PF above 1.1 after stress costs, and profits that do not depend on one or two days. **What would make us stop:** a drawdown beyond 60R (worse than anything since 2017), or a sustained stretch like 2010–16 in which the VR-filtered trades do no better than the unfiltered momentum trades (i.e. the filter stops working). The bootstrap assumes trades are independent, so real losing runs cluster more than the table shows.

Discussion

Strength of evidence. Over 2010–2026 the strategy made +623R on 4,060 trades (+0.15R per trade, t 3.77 in R, PF 1.24 in dollars). The part that could not have influenced its selection, 2023 to Aug 2026, is 1,001 trades at +0.28R (t 3.22 in R, PF 1.34). The holdout alone beat 99 of 100 no-edge simulated markets. By the campaign's rules, that is as good as untouched evidence gets.

How to read the dollar curve. Dollar P&L on one contract grows with NQ's price and volatility. 2020 was +57R but −\$5.3k, because a few large-risk COVID-era losses outweighed many small-risk wins. In R the strategy has been positive in every calendar year since 2017. The official verdicts use dollars; the R view explains them.

Where the P&L comes from. About 67% of trades hit the stop (average −1.07R). The rest are held to the close and average +2.66R. After 2022 the short side (+0.40R per trade) earned more than the long side (+0.16R). In the holdout the long-only version had PF 1.05, compared with 1.31 for the short-only version. In 2026 the pattern reversed (long 1.48, short 1.07). The frozen strategy trades both sides, so no side choice is needed. This is noted because each side on its own is noisy.

Quiet days work best. Split by stop size relative to price (ranked within each year), entries on the quietest third of days averaged +0.38R (+0.60R after 2022). Mid- and high-volatility entries were close to zero. The strategy works best when a steady trend develops in a calm market, not in panics. High-volatility periods such as March 2020 (−33.6R in one month) are where it is most exposed. This is descriptive only.

Management. The frozen rule (1× ATR stop, hold to close) is the simplest in the library and was chosen in discovery. Looking back, it was also the best stop/exit variant in the holdout. Treat that as luck, not as a reason to expect it to stay best. Every variant that lets winners run was profitable after the freeze.

Portfolio fit. H41 and H20 are the same kind of trade: continuation after a strong, persistent intraday move. Put them in one momentum risk budget, not two. If only one is traded, H41 has the better post-freeze record (H20 lost \$52.6k in 2026, while H41 made \$56.7k).

Recommendation

Recommendation: FIRST CANDIDATE FOR FORWARD TESTING. Run a small, fixed-risk pilot next to the true-forward paper record. Do not trade full size yet. H41 is the only one of the 25 frozen strategies with clear profits in every test after its selection period, on a large sample. Untouched holdout 2023–25: +\$143.4k, PF 1.31, t 2.43, null p 0.010. 2026 to date: +\$56.7k, PF 1.45. The edge is thin, though: PF 1.2–1.4, a 30% win rate, and most of the profit comes from a few big trend days. It also barely made money from 2010 to 2016. Plan for roughly half the recent edge.

Decision requested: approve H41 as the **first strategy to take into forward testing**, as a small pilot, not a full allocation.

Pilot design. Trade MNQ (micro) contracts, sized so every trade risks a fixed dollar amount: contracts = risk budget ÷ (ATRh × \$2), rounded down, with no trade if the result is below one. Set the risk per trade so that the bootstrap 99th-percentile 12-month drawdown (about 60R) is no more than 5% of the capital assigned. Example: \$250 per trade means a \$15k 99%-case drawdown, so at least \$300k assigned. The frozen rules must be followed exactly: no discretion and no skipped signals, because the edge depends on the few big days.

Alternative if no live exposure is wanted: forward-paper only, via forward/run_forward.py, with the same gates. At about 5 trades a week the first gate is reached in about 7 months.

Conditions, limits and kill criteria

- Scale-up gate: at least 150 true-forward trades with an average of +0.08R or better, net positive after stress costs, and no drawdown beyond 46R (the bootstrap 95th percentile). Then risk per trade may double. Full size needs 12 months forward at +0.08R or better.
- Kill switch (stop and review): a drawdown beyond 60R (the bootstrap 99th percentile, above anything since 2017), or a rolling 300-trade average below 0R. Even with a real +0.12R edge the 60R line is crossed in about 1 year in 20; that false-alarm rate is the price of the protection.
- Regime watch: every quarter, compare the forward trades with the unfiltered momentum trigger. If the VR filter stops adding value for two consecutive quarters, treat that as the start of a 2010–16-type regime.

- Risk budget: H41 and H20 (and the short-only H41 #15) share one momentum sleeve, because their daily correlation is 0.5–0.65.
- No parameter, filter or management changes. Every observation in this report is descriptive. Any change would be a new hypothesis, preregistered and tested on future data only.

Caveats

- Backtests use 1-minute bars with pessimistic same-bar handling and fixed slippage. They do not model queue position, fast-market slippage beyond 2 ticks per side (stress case), or exchange outages.
- The 2026 sample (Jan–10 Aug) is short. The campaign's own record rates it low-power (cohort null p 0.26).
- The R-based view and the parameter/management grids in this report were computed after the holdout and 2026 results were known. They are diagnostic only and must not be used to modify or re-select the strategy. The preregistered process says the only valid new evidence now comes from the true-forward record starting 24 Sep 2026.
- This is a quantitative research assessment of a systematic strategy, not personalised investment advice. Capital allocation decisions rest with the committee.

Appendix A: frozen specification (verbatim)

H41 (#19, long + short)

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Appendix B: provenance

Item	Value
Final cohort freeze	freezes/FINAL_COHORT_FREEZE.json · sha256 f5666dac728aa2539b2f8223309012914d71923ab71f3363e9e035344ace8f03
Frozen (UTC)	2026-09-22T23:21:51.429326+00:00
Discovery freeze	sha256 0343a974953d53dc026fd1f7ef5c124c8985213b132fba5df4f8a986fc85b2c9
Master record	results/V3_MASTER_RESEARCH_RECORD.md
Sealed holdout / audit trades	results/holdout/trades.parquet · results/audit2026/trades.parquet
This report	reports/H41_Variance_Ratio_Trending_Momentum_Research_Report.pdf, built from trades regenerated with the frozen specs (exact match with the sealed records) + stored discovery/confirmation evidence
Known erratum	results/audit2026/ERRATUM_2026-09-22.json (X3 null p only; does not affect this report)